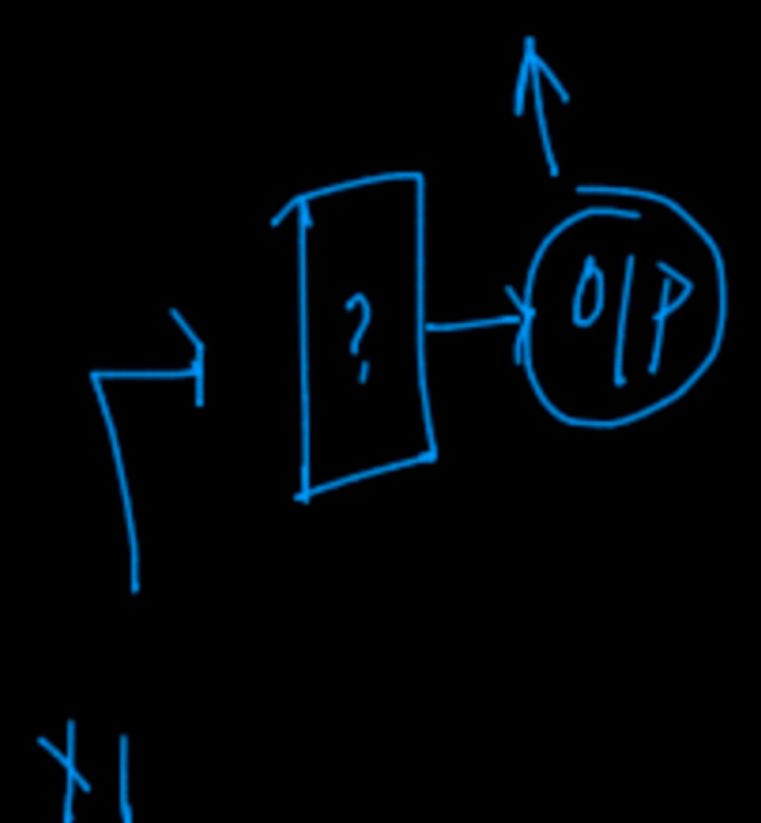
ROC and AUC

Receiver operating characteristics

Area under curve

Step1

Store prob-score and sort in decreasing



X	Y	$\hat{Y}$	$\hat{Y}_{0.95}$		$\hat{Y}_{0.92}$	$\hat{Y}_{0.76}$
X1	1	0.95	1	✓	1	1
X2	1	0.92	0	×	1	1
X3	0	0.80	0	✓	0	1
X4	1	0.76	0	×	0	1
X5	1	0.71	0	×	0	0

Step2

$\tau_1 = 0.95$

if $\tau \geq 0.95 \rightarrow 1$

else $\rightarrow 0$

	0	1
0		3
1		1

$TPR \rightarrow \frac{1}{4} \Rightarrow 0.25$ ✓

$FPR \rightarrow 0.75$ ✓

(TPR, FPR)

(TPR, FPR)

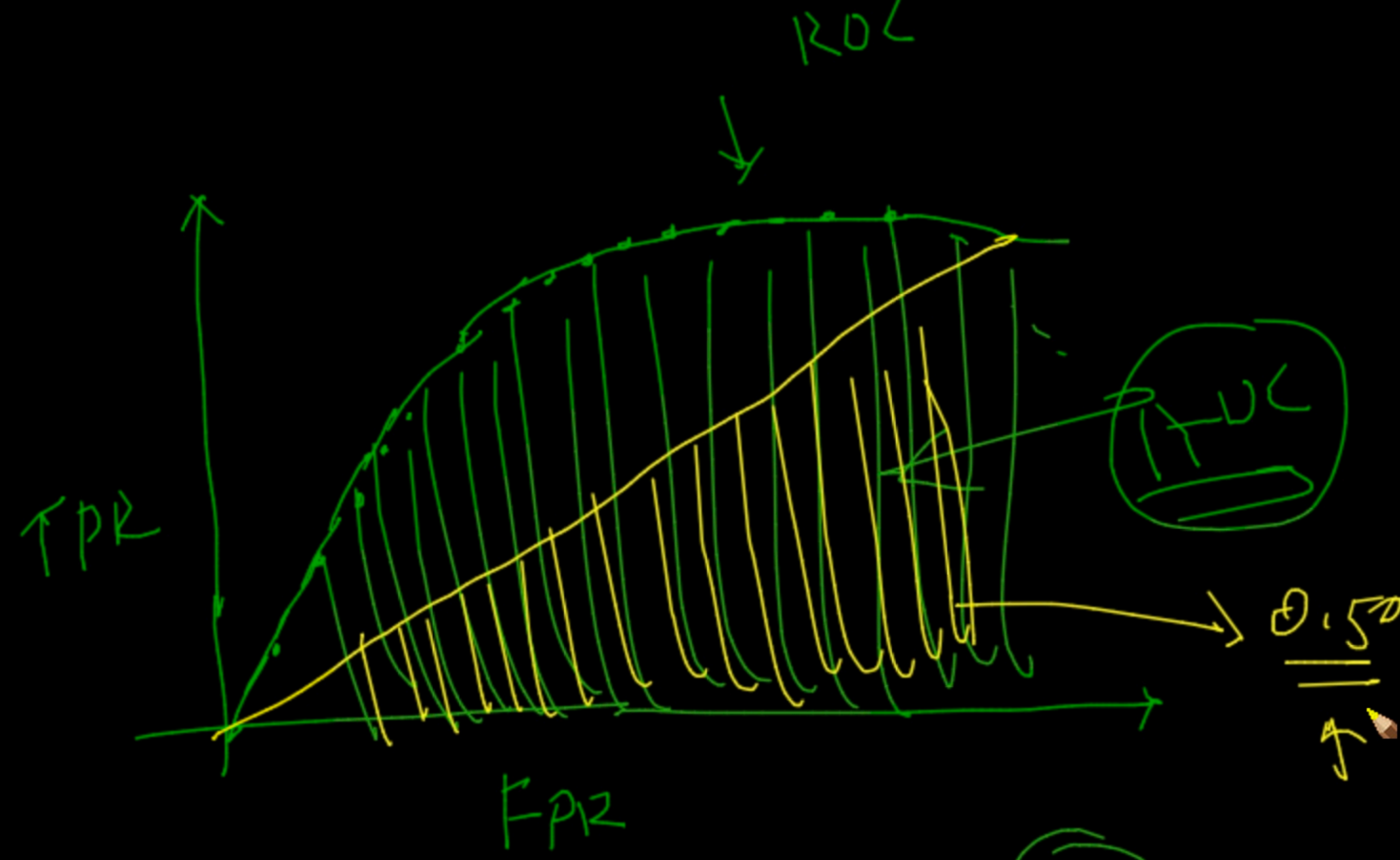
(TPR_1, FPR_1)

(TPR_2, FPR_2)

$\vdots$

(TPR_n, FPR_n)

plot it →



model → AUC, ROC → (0 – 1)

terrible

0.5

'v good'

(VI) Log-loss

↑

Binary classification

M1 M2

Best

↓

Acc → ?

↓

Prob-score → log-loss

X	Y	$\hat{Y}$
X1	1	0.9
X2	1	0.6
X3	0	0.1
X4	0	0.4

0.0.457

$\{\log(0.9)\} * 1 + (1-p) - -$

$-\log(0.6)$

0.122

0.122

$-\log(0.1) \to + (1)\log(0.9)$

0.045

0.122

log loss ⇒

X1

$$\text{log-loss} = -\frac{1}{n}\sum_{i=1}^{n} \log(p_i) * y_i + (1-y_i) * \log(1-p_i)$$

3 / 3

X	Y_{Actual}	$Y_{Predicted}$	error
X1	220	250 ✓	(-30)
X2	250	210	(-40)

→ 30

40

70

SD